

Residency Project:

Advanced Data Mining for Data-Driven Insights and Predictive Modeling

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(MSCS-634-M20) Advanced Big Data and Data Mining

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1. Introduction

This report consolidates a four-part data mining project built on the Clickstream Data for Online Shopping dataset (UCI Machine Learning Repository, ID 553). The dataset records browsing activity from an online store selling maternity clothing over five months in 2008: 165,474 individual clicks across 14 attributes, which we aggregated to 24,026 browsing sessions.

The project moved through four stages, each asking a different question of the same data:

- Deliverable 1: Data cleaning and exploration: is the dataset usable, and what does it actually contain?
- Deliverable 2: Regression: which products attract customer attention, and does price play a part?
- Deliverable 3: Classification, clustering, and pattern mining: can we predict engagement early, what types of shopper exist, and which products get browsed together?
- Deliverable 4: Synthesis, what do these four analyses say together, what are the practical recommendations, and what ethical considerations apply to the data and the conclusions drawn from it?

We chose this dataset because it supports all four required analyses from a single, coherent source, and because.

2. Dataset and Data Presentation

2.1.Dataset Description

Each record in the raw data is a single click: the product viewed, its price, its color, where its photo appeared on the page, the page depth reached, the date, and the country the visitor's IP resolved to. There is no purchase outcome in the data, only browsing behavior.

2.2.Cleaning Steps

- Six categorical attributes (country, category, color, photo location, model photography, price band) were stored as integer codes. Left as numbers, regression models would have imposed an artificial ordering on unordered categories, and distance-based methods (K-Means, k-NN) would have computed meaningless distances between category labels. All were recoded to labels using the official data dictionary; page_number was deliberately kept numeric, since page depth is a genuine ordinal quantity.
- The year column was constant at 2008 and dropped for carrying no information.
- Missing values existed despite the source documentation implying a complete dataset: country code 12 denotes an IP address that could not be geolocated. This was not detectable with a standard null check, it required reading the data dictionary. The affected 210 records were retained (their category, price, and page-depth values are still valid) with country set to missing and a country_known flag added.
- The date, split across three separate columns, was reassembled and validated.

2.3.Unit of Analysis

The raw data is one row per click, but the project's questions are about sessions and shoppers. Left at click level, a visitor who clicked 50 times would count 50 times toward any pattern, while a visitor who clicked twice would count twice, heavy browsers would dominate every result. Clicks were therefore aggregated to one row per session (24,026 sessions from 165,474 clicks), while the click-level table was retained for the market-basket analysis, where each session is itself the natural transaction.

2.4.Outlier Handling

Session length and other session-level aggregates were assessed for outliers using both the IQR rule and z-scores. All sessions were retained: an unusually long session is not a measurement error but a real, informative instance of the behavior the project is trying to understand, and web session lengths are naturally heavy-tailed. Extreme sessions were flagged and a log transform was prepared for the scale-sensitive clustering step, rather than deleting them.

2.5.A Structural Finding That Shaped the Rest of the Project

Before building any model, we checked whether price could actually be predicted from the product code. It could not, in the sense that mattered: every one of the 217 distinct products in the store has exactly one fixed price, and in fact every product attribute: color, screen position, photo style, category, is fixed per product. The 165,474 click records therefore contain only 217 distinct feature combinations, repeated thousands of times. Predicting price from the product code would have been recovering a lookup table, not a genuine model, and training a click-level model on it would have leaked the answer between train and test splits (the same product appears in both). This finding is the reason Regression step was reframed around product-level attention rather than click-level price.

3. Regression Modeling: What Drives Product Attention

Because price is fixed per product, the useful regression question runs in the opposite direction from the one originally planned: which products attract customer attention (measured as total clicks received), and does price play a part? This reframing also required moving the unit of analysis from clicks to products (217 rows), since every click on a given product carries identical features.

3.1.Feature Engineering

Four engineered features were added to the raw product attributes: price_band (a quartile bucket, capturing non-linear price effects a single numeric term would miss), is_sale_item, is_top_row (isolating vertical screen position), and clicks_per_session (used for interpretation only, not as a predictor, since it is calculated from the target and would leak).

3.2.Models and Results

With only 217 observations and roughly 28 encoded features, an unregularized model has enough freedom to fit noise, a ratio that makes regularization genuinely necessary rather than optional. Four models were compared using 5-fold cross-validation:

Model	CV R^2	R^2 Std Dev	CV MSE	CV RMSE
Linear Regression	0.1773	0.2490	275692.7182	515.7351
Ridge (alpha=20)	0.3157	0.1330	238560.3801	477.1335
Lasso (alpha=20)	0.3200	0.0984	242247.4804	478.9623
Random Forest (depth=2)	0.4571	0.0851	189352.5080	425.7796

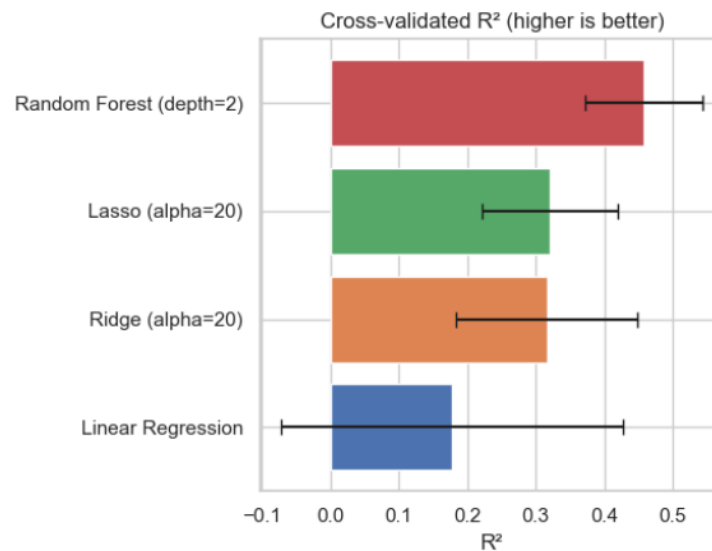


Figure 1. Cross-validated R^2 by model, with fold-to-fold standard deviation. The depth-limited Random Forest performs best; plain Linear Regression overfits the 217-row sample.

The ordering across all four models tells a single consistent story: with this little data, constraining the model matters more than making it flexible. Plain Linear Regression performs worst and has the widest fold-to-fold spread, the signature of overfitting. Ridge and Lasso both improve substantially on it by shrinking coefficients. The depth-limited Random Forest performs best of all, but only when kept shallow; an unrestricted forest performs worse than Ridge.

3.3.Key Findings: Higher Price, More Attention

Higher-priced products receive more attention, not less, a positive correlation of about 0.26 between price and click count, the opposite of the intuitive expectation that shoppers gravitate toward cheaper items. The relationship is also not monotonic: attention rises from budget to mid-low price bands, dips at mid-high, then peaks again at premium. Category matters more than price overall, skirts and trousers draw roughly twice the attention of sale items, and the sale category is the least-viewed of the four, which sits oddly next to the finding that premium prices draw more interest. Both results point away from discounting as the main driver of engagement in this store. Color contributes little; Lasso eliminated 27 of 34 encoded features entirely, most of them color indicators.

3.4.Limitations

- 217 observations is a small sample; cross-validation standard deviations of ± 0.09 to ± 0.25 mean the gap between models is real but not precisely measured.
- The dataset records browsing only, not purchases, attention is not the same as revenue, and none of these findings should be read as a sales recommendation without further validation.

- The relationships identified are correlational. Premium products may attract attention because of their price, or because they are also the newest, best-photographed, or most prominently placed items; this data cannot separate those explanations.

4. Classification, Clustering, and Pattern Mining

Deliverable 2 asked which products attract attention. Deliverable 3 shifted the focus to visitors, asking three separate questions: can engagement be predicted early in a session, what distinct types of shopping session exist, and which products get browsed together?

4.1. Classification: Predicting Engagement from the First Two Clicks

The target was whether a session reached page 3 or deeper ("deep" vs. "shallow"), predicted using only the visitor's first two clicks, what they landed on, its price and color, where the photo sat, plus country and weekday. Framing the problem around opening clicks only, rather than the whole session, was a deliberate choice: predicting depth from a complete session is circular, since a long session is deep almost by definition. Using only the opening two clicks makes it a genuine prediction problem, and a practically useful one, a retailer could act on it while the visitor is still on the site. This left 18,984 of 24,026 sessions with a usable second click; the class split was close to even (majority-class baseline accuracy 0.544).

Three classifiers were compared, with the Decision Tree tuned via GridSearchCV over max_depth, min_samples_leaf, and criterion, scored on F1 rather than accuracy so the search would not simply reward favoring the majority class:

Model	Accuracy	F1	ROC-AUC
Decision Tree (tuned)	0.6355	0.5667	0.6693
k-NN (k=25)	0.6286	0.5387	0.6706
Naive Bayes	0.5959	0.4790	0.6288

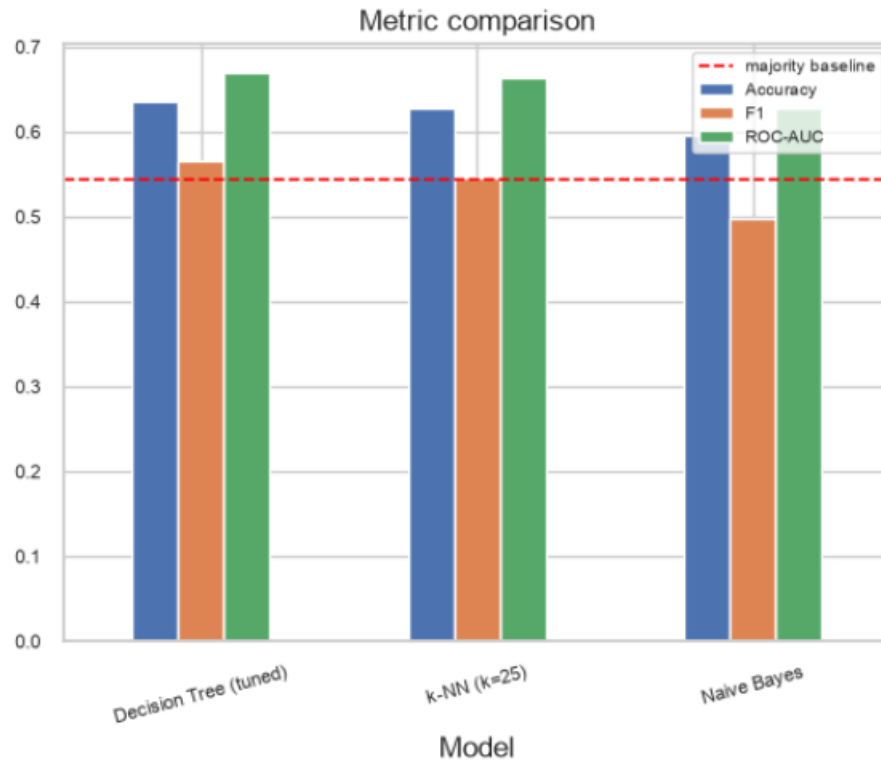


Figure 2. All three classifiers beat the majority-class baseline (0.544), with the tuned Decision Tree performing best.

All three models beat the baseline, so the first two clicks carry real, if modest, signal about eventual engagement. The tuned tree catches about 73% of shallow sessions but only 52% of deep ones, a meaningful gap, since in a real deployment missing an engaged visitor (a false negative) plausibly costs more than an unnecessary prompt to a casual browser (a false positive). A production version of this model would reasonably be tuned toward recall, at the cost of accuracy, and its decision threshold lowered accordingly.

4.2.Clustering: Four Types of Shopping Sessions

K-Means was applied to standardized, session-level features (with a log transform on click count to control for the heavy right skew identified in Deliverable 1). The silhouette score was highest at $k=2$, but that solution only separates "short" and "long" sessions, a distinction click

count alone already captures, so the clustering added no new information. $k=4$ was used instead, trading a lower silhouette score for a solution that separates sessions on two dimensions, engagement level and price level, rather than one.

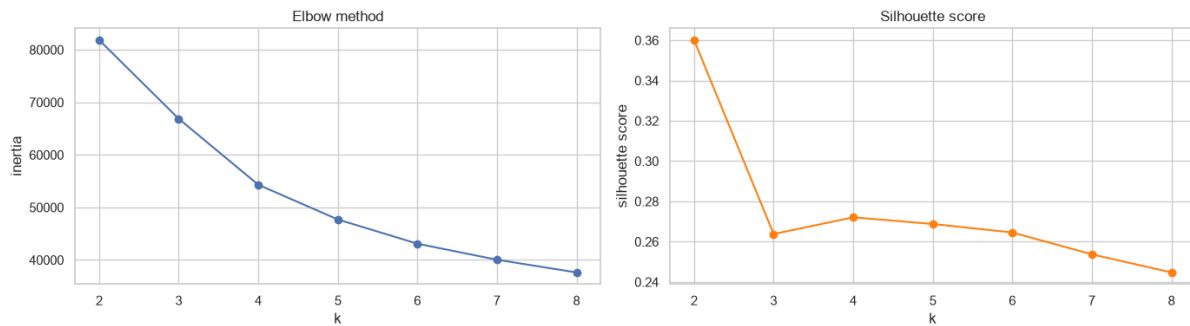


Figure 3. Elbow (inertia) and silhouette score across $k=2-8$, from the Deliverable 3 notebook. Silhouette peaks at $k=2$, but that split is redundant with click count alone.

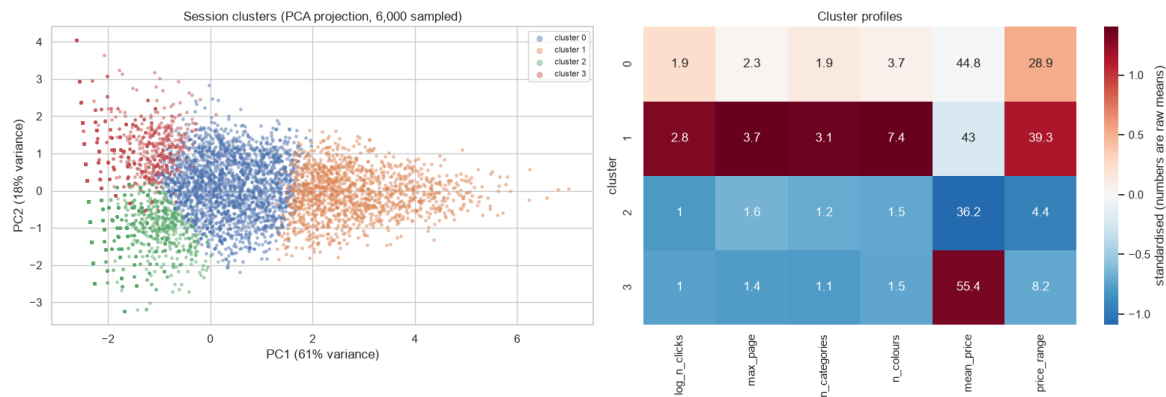


Figure 4. PCA projection of the four K-Means clusters (left) and their behavioral profile heatmap (right), taken directly from the Deliverable 3 notebook.

- Cluster 2 — Budget bouncers (~5,750 sessions): about 2 clicks, lowest average price (~\$36), one category. Arrive, view something cheap, leave.
- Cluster 3 — Premium bouncers (~4,600 sessions): about 2 clicks, highest average price (~\$55), one category. The same brief visit, but at the opposite end of the price range.
- Cluster 0 — Moderate browsers (~8,400 sessions, the largest group): about 6 clicks across 2 categories at mid-range prices.

- Cluster 1 — Deep explorers (~5,300 sessions): about 18 clicks, 3 categories, 7 colors, and the widest price range, highly engaged.

The most actionable result is that the two brief-visit clusters are not the same segment. A conventional bounce-rate report would merge budget and premium bouncers into one "left after two clicks" group, but they are looking at opposite ends of the price range and plausibly need different treatment, price reassurance for one, quality or delivery information for the other. This is the concrete, practical reason the lower-silhouette k=4 solution was preferred over the statistically cleaner k=2 split.

4.3. Association Rule Mining: What Gets Browsed Together

Each session was treated as a transaction, or "basket", of the categories and price tiers viewed during it (24,026 transactions, 8 distinct items, averaging 4.3 items per session). This is a natural fit for the dataset: unlike most tabular data used for rule mining, sessions are already baskets, requiring no artificial discretization. Apriori was run with a minimum support of 0.03, and rules were filtered to lift above 1.05 and confidence at or above 0.4, yielding 274 rules from 92 frequent itemsets.

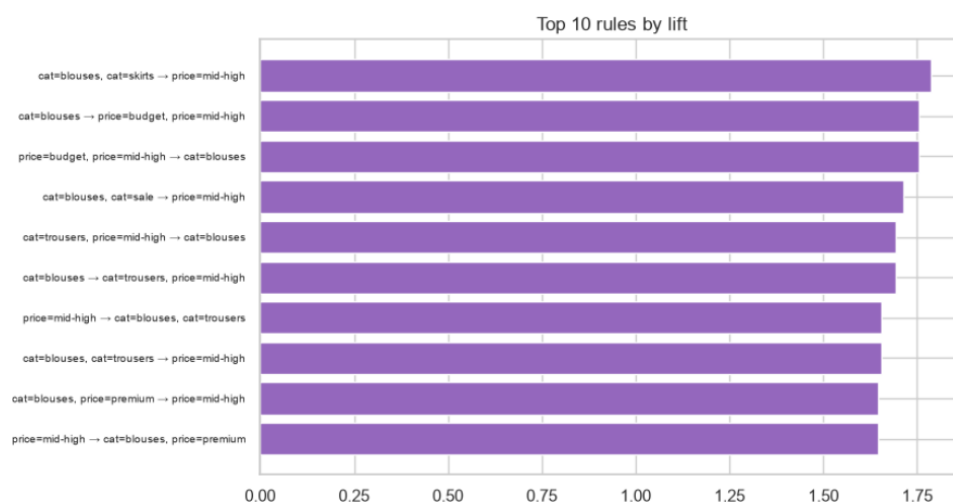


Figure 5. Top association rules by lift, mostly linking garment categories with the mid-high price tier.

Two patterns dominate the top rules. First, shoppers browse across categories rather than within a single one: rules linking blouses with trousers and skirts recur at lift 1.65–1.79, indicating visitors are assembling outfits rather than searching for one item type. Second, shoppers mix price tiers within a single session, budget items co-occur with mid-high items at lift 1.76, which argues against treating "budget shopper" and "premium shopper" as fixed customer types, at least within one browsing session.

5. Practical Recommendations

Combining the findings across all three deliverables suggests several concrete, actionable directions for the retailer:

- Lead merchandising with prominent placement of higher-priced skirts and trousers rather than the sale range. Both the regression result (positive price–attention correlation, sale items least-viewed) and the clustering result (a dedicated, sizable premium-bouncer segment) point the same way, discounting is not clearly the primary driver of engagement in this store.
- Build cross-category recommendations ("complete the outfit") rather than more-of-the-same suggestions. The association rules consistently link garment categories rather than items within one category, so a recommender that only shows more blouses to someone viewing blouses misses the dominant real behavior.
- Treat the two brief-visit segments differently. Budget bouncers and premium bouncers both leave after roughly two clicks, but a single retention message aimed at "bouncers" as a group would misread half of them; price-tier-aware messaging is better supported by the clustering result.

- Target deep explorers (about 22% of sessions) for recommendation and loyalty features. This segment already browses widely across categories and colours, making it the natural audience for cross-sell and personalization investment.
- Use early-click signals (first two clicks) as a low-confidence trigger for real-time intervention, a chat prompt, recommendation, or offer, for visitors predicted to disengage, tuning the classifier toward recall since missing an engaged visitor is more costly than an unnecessary prompt.

Every recommendation above should be read against the central limitation of the dataset: it records browsing, not purchasing. No result here can be validated against revenue without additional data, and all recommendations should be piloted and measured against actual conversion before being treated as settled.

6. Ethical Considerations

6.1. Geolocation Without Explicit Consent

The country attribute is derived from the visitor's IP address rather than from anything the visitor provided directly. This is a common but non-trivial practice: it collects locational information passively, without an explicit consent step, and 2008-era e-commerce sites of this kind typically did not disclose IP-based geolocation clearly to users. We addressed this by using country only in aggregate (e.g., the finding that traffic is 81% domestic) and never to identify or target individual visitors, and by flagging the 210 records where geolocation failed (country code 12) rather than silently dropping or guessing at them.

6.2. Temporal and Regular Context

The data was collected in 2008, well before the EU's General Data Protection Regulation (GDPR, effective 2018) and most modern data-privacy law. Passive collection of IP-based location and detailed clickstream behavior wit, legal requirements today. This project treats the dataset as a historical artifact for methodological practice, not as a template for how a retailer should collect data now; any real deployment of these techniques on live customer data today would need to be built around informed consent, data minimization, and a lawful basis for processing under current regulation.

6.3.Fairness and Bias in the Models

The classification and clustering models use behavioral and product features (clicks, price, category, screen position) rather than demographic attributes, which limits direct disparate treatment by protected characteristics such as gender or age, none are present in the dataset. However, indirect bias is still possible: country acts as a rough proxy for nationality, and the dataset is heavily skewed toward one country (about 81% of sessions are domestic to Poland), meaning any pattern involving the remaining 19% of international traffic is drawn from a much smaller, less reliable sample. Conclusions about "international shoppers" as a group should be treated with more caution than conclusions about the domestic majority, and we avoided building or reporting any model that segments or targets specifically by country for this reason.

6.4.Generalizability

The findings in this report describe one retailer, in one country, over five months in 2008. They should not be generalized to online shopping behavior broadly, to other retail categories, or to the present day without independent validation. This is a modeling and methodology exercise rather than a basis for a live business decision.

7. Conclusion

Across four deliverables, this project moved from raw, undocumented click logs to a set of specific, defensible findings: which products earn attention and why (Deliverable 2), which visitors are likely to stay engaged and what distinct shopper types exist (Deliverable 3), and what those results mean once the sensitive nature of a maternity retailer's data is taken seriously (Deliverable 4). The most consistent theme across every stage is the value of checking assumptions before modeling, the discovery that price is fixed per product reshaped the entire regression approach, the discovery that $k=2$ clustering collapsed two economically different groups reshaped the segmentation, and the recognition that this is a maternity retailer reshaped how every subsequent finding should be used.

8. References

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(Apriori/association_rules implementation.)